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B.Tech. Degree VI Semester Examination April 2019

ME 15-1605 CAD/CAM (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

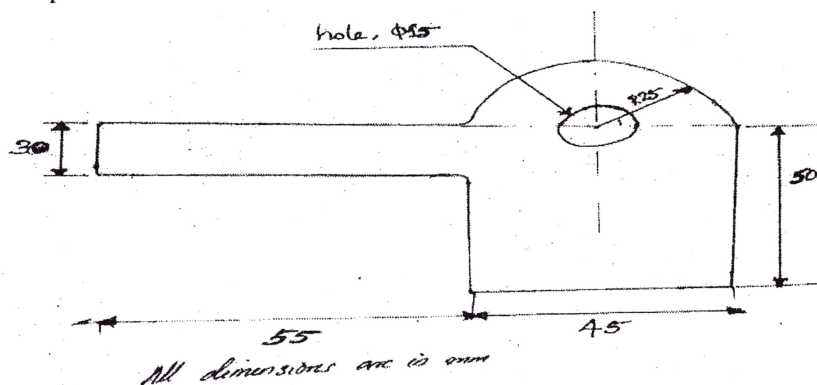
- I. (a) Distinguish between absolute positioning and incremental positioning?
 (b) Define FEM?
 (c) Define the term NC, CNC, DNC and CIM?
 (d) What are different types of statement used in APT?
 (e) What is CAPP? What are the challenges for CAPP?
 (f) What is flow line balancing and line balancing?
 (g) Define work volume of Robot?
 (h) Classify the robot based on its configuration?
 (i) Differentiate between manual part programming and computer part program?
 (j) Differentiate between Miscellaneous and preparatory functions used in manual part program?



PART B

(4 × 10 = 40)

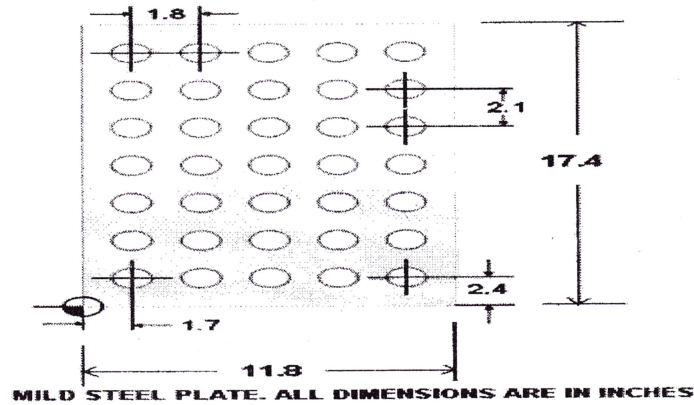
- II. (a) Explain the steps in data exchange between CAD and CAM? (8)
 (b) What is Detroit type automation? (2)
- OR**
- III. (a) What are the merits and demerits of CAD? (2)
 (b) Explain the different levels of Automation? (8)
- IV. Write a complete APT part program to machine the outline of the geometry and drill a hole as shown in the figure. The component is 5mm thick. The end mill used is 10mm in diameter. Assume the spindle speed as 955 rpm? (10)



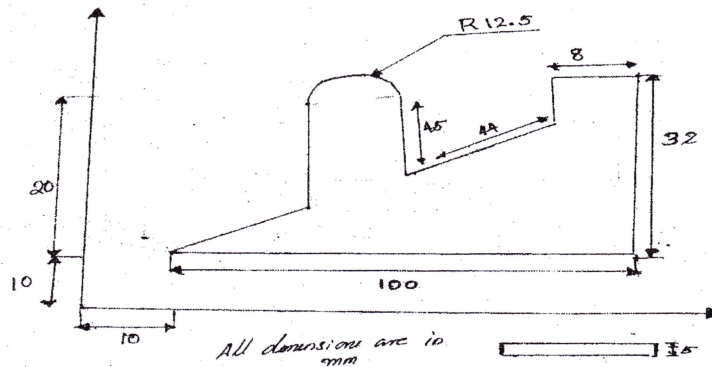
OR

(P.T.O.)

- V. Write an efficient CNC part program to drill 35 holes of diameter 0.5 inch each in a machine component as shown in figure. The raw material used is mild steel plate of 0.4 inch thickness. Explain the important function used in CNC code? (10)



- VI. Write a detailed APT geometric statement to drive the outline of the CAM as shown in figure with cutter diameter of 65mm, cutting speed of 950 rpm and feed rate is 8mm/min? (10)



OR

- VII. Define and differentiate closed loop control and open loop control? (10)
- VIII. (a) How do the experts evolve out of Artificially Intelligent system? (8)
- (b) List out the application of robots? (2)

OR

- IX. (a) Discuss the structure of Robotic system? (5)
- (b) Write a short note on control systems in Robot? (5)
